

- Ilya Loshchilov and Frank Hutter. SGDR: stochastic gradient descent with warm restarts. In *5th International Conference on Learning Representations, ICLR 2017, Toulon, France, April 24-26, 2017, Conference Track Proceedings*. OpenReview.net, 2017. URL <https://openreview.net/forum?id=Skq89Scxx>.
- Kairong Luo, Haodong Wen, Shengding Hu, Zhenbo Sun, Zhiyuan Liu, Maosong Sun, Kaifeng Lyu, and Wenguang Chen. A Multi-Power law for loss curve prediction across learning rate schedules. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. URL <https://openreview.net/forum?id=KnoS9XxI1K>.
- Michael McCloskey and Neal J Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of learning and motivation*, volume 24, pp. 109–165. Elsevier, 1989.
- Todor Mihaylov, Peter Clark, Tushar Khot, and Ashish Sabharwal. Can a suit of armor conduct electricity? a new dataset for open book question answering. In Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun’ichi Tsujii (eds.), *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pp. 2381–2391, Brussels, Belgium, oct 2018. Association for Computational Linguistics. doi: 10.18653/v1/D18-1260. URL <https://aclanthology.org/D18-1260/>.
- Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, Michal Guerquin, Hamish Ivison, Pang Wei Koh, Jiacheng Liu, Saumya Malik, William Merrill, Lester James V. Miranda, Jacob Morrison, Tyler Murray, Crystal Nam, Valentina Pyatkin, Aman Rangapur, Michael Schmitz, Sam Skjonsberg, David Wadden, Christopher Wilhelm, Michael Wilson, Luke Zettlemoyer, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. 2 OLMo 2 furious. *CoRR*, abs/2501.00656, 2025. doi: 10.48550/ARXIV.2501.00656. URL <https://doi.org/10.48550/arXiv.2501.00656>.
- OpenAI. GPT-4 technical report. *CoRR*, abs/2303.08774, 2023. doi: 10.48550/ARXIV.2303.08774. URL <https://doi.org/10.48550/arXiv.2303.08774>.
- Guilherme Penedo, Quentin Malartic, Daniel Hesslow, Ruxandra Cojocaru, Hamza Alobeidli, Alessandro Cappelli, Baptiste Pannier, Ebtesam Almazrouei, and Julien Launay. The RefinedWeb dataset for Falcon LLM: outperforming curated corpora with web data only. In Alice Oh, Tristan Naumann, Amir Globerson, Kate Saenko, Moritz Hardt, and Sergey Levine (eds.), *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*, 2023. URL http://papers.nips.cc/paper_files/paper/2023/hash/fa3ed726cc5073b9c31e3e49a807789c-Abstract-Datasets_and_Benchmarks.html.
- Guilherme Penedo, Hynek Kydlíček, Vinko Sabolcec, Bettina Messmer, Negar Foroutan, Amir Hossein Kargaran, Colin Raffel, Martin Jaggi, Leandro von Werra, and Thomas Wolf. Fineweb2: One pipeline to scale them all - adapting pre-training data processing to every language. *CoRR*, abs/2506.20920, 2025. doi: 10.48550/ARXIV.2506.20920. URL <https://doi.org/10.48550/arXiv.2506.20920>.
- Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. WinoGrande: An adversarial winograd schema challenge at scale. In *The Thirty-Fourth AAAI Conference on Artificial Intelligence, AAAI 2020, The Thirty-Second Innovative Applications of Artificial Intelligence Conference, IAAI 2020, The Tenth AAAI Symposium on Educational Advances in Artificial Intelligence, EAAI 2020, New York, NY, USA, February 7-12, 2020*, pp. 8732–8740. AAAI Press, 2020. doi: 10.1609/AAAI.V34I05.6399. URL <https://doi.org/10.1609/aaai.v34i05.6399>.
- Sunny Sanyal, Atula Tejaswi Neerkaje, Jean Kaddour, Abhishek Kumar, and sujay sanghavi. Early weight averaging meets high learning rates for LLM pre-training. In *First Conference on Language Modeling*, 2024. URL <https://openreview.net/forum?id=IA8CWtNkUr>.

- Maarten Sap, Hannah Rashkin, Derek Chen, Ronan Le Bras, and Yejin Choi. Social IQa: Commonsense reasoning about social interactions. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan (eds.), *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 4463–4473, Hong Kong, China, November 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1454. URL <https://aclanthology.org/D19-1454/>.
- KaShun SHUM, Yuzhen Huang, Hongjian Zou, dingqi, YiXuan Liao, Xiaoxin Chen, Qian Liu, and Junxian He. Predictive data selection: The data that predicts is the data that teaches. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=tTVYR82Iz6>.
- Dan Su, Kezhi Kong, Ying Lin, Joseph Jennings, Brandon Norick, Markus Kliegl, Mostofa Patwary, Mohammad Shoeibi, and Bryan Catanzaro. Nemotron-CC: Transforming Common Crawl into a refined long-horizon pretraining dataset. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (eds.), *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, pp. 2459–2475. Association for Computational Linguistics, 2025. URL <https://aclanthology.org/2025.acl-long.123/>.
- Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. CommonsenseQA: A question answering challenge targeting commonsense knowledge. In Jill Burstein, Christy Doran, and Tamar Solorio (eds.), *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pp. 4149–4158, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1421. URL <https://aclanthology.org/N19-1421/>.
- Meituan LongCat Team, Bayan, Bei Li, Bingye Lei, Bo Wang, Bolin Rong, Chao Wang, Chao Zhang, Chen Gao, Chen Zhang, Cheng Sun, Chengcheng Han, Chenguang Xi, Chi Zhang, Chong Peng, Chuan Qin, Chuyu Zhang, Cong Chen, Congkui Wang, Dan Ma, Daoru Pan, Defei Bu, Dengchang Zhao, Deyang Kong, Dishan Liu, Feiye Huo, Fengcun Li, Fubao Zhang, Gan Dong, Gang Liu, Gang Xu, Ge Li, Guoqiang Tan, Guoyuan Lin, Haihang Jing, Haomin Fu, Haonan Yan, Haoxing Wen, Haozhe Zhao, Hong Liu, Hongmei Shi, Hongyan Hao, Hongyin Tang, Huantian Lv, Hui Su, Jiacheng Li, Jiahao Liu, Jiahuan Li, Jiajun Yang, Jiaming Wang, Jian Yang, Jianchao Tan, Jiaqi Sun, Jiaqi Zhang, Jiawei Fu, Jiawei Yang, Jiaxi Hu, Jiayu Qin, Jingang Wang, Jiyuan He, Jun Kuang, Junhui Mei, Kai Liang, Ke He, Kefeng Zhang, Keheng Wang, Keqing He, Liang Gao, Liang Shi, Lianhui Ma, Lin Qiu, Lingbin Kong, Lingtong Si, Linkun Lyu, Linsen Guo, Liqi Yang, Lizhi Yan, Mai Xia, Man Gao, Manyuan Zhang, Meng Zhou, Mengxia Shen, Mingxiang Tuo, Mingyang Zhu, Peiguang Li, Peng Pei, Peng Zhao, Pengcheng Jia, Pingwei Sun, Qi Gu, Qianyun Li, Qingyuan Li, Qiong Huang, Qiyuan Duan, Ran Meng, Rongxiang Weng, Ruichen Shao, Rumei Li, Shizhe Wu, and Shuai Liang. LongCat-Flash technical report. *CoRR*, abs/2509.01322, 2025. doi: 10.48550/ARXIV.2509.01322. URL <https://doi.org/10.48550/arXiv.2509.01322>.
- Teknium. OpenHermes 2.5: An open dataset of synthetic data for generalist llm assistants, 2023. URL <https://huggingface.co/datasets/teknium/OpenHermes-2.5>.
- Changxin Tian, Jiapeng Wang, Qian Zhao, Kunlong Chen, Jia Liu, Ziqi Liu, Jiaxin Mao, Wayne Xin Zhao, Zhiqiang Zhang, and Jun Zhou. WSM: decay-free learning rate schedule via checkpoint merging for LLM pre-training. *CoRR*, abs/2507.17634, 2025. doi: 10.48550/ARXIV.2507.17634. URL <https://doi.org/10.48550/arXiv.2507.17634>.
- Howe Tissue, Venus Wang, and Lu Wang. Scaling law with learning rate annealing. *CoRR*, abs/2408.11029, 2024. doi: 10.48550/ARXIV.2408.11029. URL <https://doi.org/10.48550/arXiv.2408.11029>.
- Maurice Weber, Daniel Y. Fu, Quentin Anthony, Yonatan Oren, Shane Adams, Anton Alexandrov, Xiaozhong Lyu, Huu Nguyen, Xiaozhe Yao, Virginia Adams, Ben Athiwaratkun, Rahul Chalamala, Kezhen Chen, Max Ryabinin, Tri Dao, Percy Liang, Christopher Ré, Irina Rish, and

- Ce Zhang. Redpajama: an open dataset for training large language models. In Amir Globersons, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng Zhang (eds.), *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, 2024. URL http://papers.nips.cc/paper_files/paper/2024/hash/d34497330b1fd6530f7afd86d0df9f76-Abstract-Datasets_and_Benchmarks_Track.html.
- Kaiyue Wen, Zhiyuan Li, Jason S. Wang, David Leo Wright Hall, Percy Liang, and Tengyu Ma. Understanding warmup-stable-decay learning rates: A river valley loss landscape view. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=m51BgoqvbP>.
- Alexander Wettig, Aatmik Gupta, Saumya Malik, and Danqi Chen. QuRating: Selecting high-quality data for training language models. In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. OpenReview.net, 2024. URL <https://openreview.net/forum?id=GLGYqPwjy>.
- Alexander Wettig, Kyle Lo, Sewon Min, Hannaneh Hajishirzi, Danqi Chen, and Luca Soldaini. Organize the web: Constructing domains enhances pre-training data curation. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=boSqwdvJVC>.
- An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report. *CoRR*, abs/2412.15115, 2024a. doi: 10.48550/ARXIV.2412.15115. URL <https://doi.org/10.48550/arXiv.2412.15115>.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jian Yang, Jiayi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruizhe Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report. *CoRR*, abs/2505.09388, 2025. doi: 10.48550/ARXIV.2505.09388. URL <https://doi.org/10.48550/arXiv.2505.09388>.
- Enneng Yang, Zhenyi Wang, Li Shen, Shiwei Liu, Guibing Guo, Xingwei Wang, and Dacheng Tao. Adamerging: Adaptive model merging for multi-task learning. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024b. URL <https://openreview.net/forum?id=nZP6NgD3QY>.
- Hu Yiwen, Huatong Song, Jie Chen, Jia Deng, Jiapeng Wang, Kun Zhou, Yutao Zhu, Jinhao Jiang, Zican Dong, Yang Lu, Xu Miao, Xin Zhao, and Ji-Rong Wen. YuLan-mini: Pushing the limits of open data-efficient language model. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (eds.), *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 5374–5400, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-251-0. doi: 10.18653/v1/2025.acl-long.268. URL <https://aclanthology.org/2025.acl-long.268/>.
- Yang Zhang, Amr Mohamed, Hadi Abdine, Guokan Shang, and Michalis Vazirgiannis. Beyond random sampling: Efficient language model pretraining via curriculum learning. *CoRR*, abs/2506.11300, 2025. doi: 10.48550/ARXIV.2506.11300. URL <https://doi.org/10.48550/arXiv.2506.11300>.